Gurudatta Patil

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Technical Skills

Languages: Go (Golang), Python, TypeScript, C++, SQL, Bash
Cloud & DevOps: Terraform, Pulumi (IaC), Kubernetes Operators/Controllers, Spinnaker, Docker, GCP (GKE), AWS, CI/CD (Cloud Build, GitHub Actions), Prometheus/Grafana
Databases: PostgreSQL, MongoDB, DynamoDB, Firestore, SQL, Redis
Backend & Systems: gRPC/Protobuf, REST APIs, HTTP/DNS, Microservices, Distributed Systems, WebSockets

Experience

- **HP Inc.** Jan 2025 – Present
  Software Development Engineer II, Platform Engineering
  - Architected the cloud infrastructure backbone for the video conferencing platform in Terraform, Kubernetes, and Minio, enabling self-service environment provisioning across 3 engineering teams and cutting deployment time from 2 days to 20 minutes.
  - Engineered Sonar, a high throughput Go microservice managing 1,000+ simultaneous A/V streams; authored reference architectures in code for containerized scheduling and distributed storage adopted org-wide.
  - Migrated legacy E2E test infrastructure to a containerized, IaC managed Docker/Kubernetes stack supporting 1,000+ scenarios, building custom Kubernetes Operators to standardize checkpointing and reduce redundant test reruns.

  Software Engineer Intern May 2024 – Aug 2024
  - Delivered a Go and Python speaker diarization service with rigorous unit/integration test coverage, reducing false positives by 93% and latency by 84%.
  - Fine tuned and quantized production ML models to 96% accuracy while halving compute footprint, reinforcing reliability standards for edge deployment pipelines.
- **Razorpay** Feb 2022 – Aug 2023
  Software Development Engineer, Platform Engineering
  - Owned hybrid cloud provisioning using Golang, Spinnaker, Terraform, and Kubernetes, building custom Kubernetes Operators to establish automated CI/CD and self healing release workflows for core backend services.
  - Engineered Razorpay's OAuth service as an independent, horizontally scalable Go microservice, standardizing self-service authentication adoption across every internal product.
  - Established Prometheus and Grafana observability across the platform, driving a 40% reduction in critical on-call incidents through proactive, self healing alert design.
- **Fragments** Dec 2021 – Dec 2022
  Lead Engineer
  - Owned and delivered the complete cloud infrastructure (AWS Lambda, DynamoDB, CloudFront) for a DAO tooling platform scaling to 25k MAU, cutting infrastructure costs 60% versus an equivalent EC2 based architecture through IaC driven serverless design.
  - Managed cloud resource provisioning and cost controls single handedly across dev and production environments, keeping the platform at zero downtime with no dedicated ops headcount.
  - Scaled the engineering organization from 1 to 6 people while retaining full ownership of platform architecture and operational reliability.

Projects

- **GameAI: Cloud Native Backend Platform (Go, Terraform, GKE, Kubernetes, Docker)**
  - Architected a Go microservices backend on GKE with full Terraform managed IaC (clusters, Artifact Registry, GCS, firewall rules), supporting 100k+ concurrent users with zero-downtime rolling deployments.
  - Engineered a Google Cloud Build CI/CD pipeline that automates containerized builds and deployments, enabling self service releases without manual intervention.
  - Designed Redis backed pub/sub for cross instance delivery behind HAProxy, achieving horizontal scalability across stateless replicas.
- **Stitch: Polyglot Code Generation & IPC Framework (Go, TypeScript, Python, Rust)**
  - Built a code generation system spanning 16 language pairs, designing reusable architecture templates that scaffold typed SDKs and eliminate manual bridge writing.
  - Implemented self healing process lifecycle management, health checks, and exponential backoff auto restart in Go, mirroring production grade infrastructure resiliency patterns.

Education

- **University of Wisconsin-Madison** Aug 2023 – Dec 2024
  M.S. Computer Engineering (GPA: 4.0/4.0)
  - Coursework: Distributed Systems, System Design / Advanced Computer Architecture, Database Management Systems, Machine Learning
- **Birla Institute of Technology and Science, Pilani** Aug 2018 – Dec 2021
  B.E. Computer Engineering
  - Coursework: Operating Systems, Computer Networks, Data Structures & Algorithms, Object-Oriented Programming